BEYOND THE BASIC SCRIPT

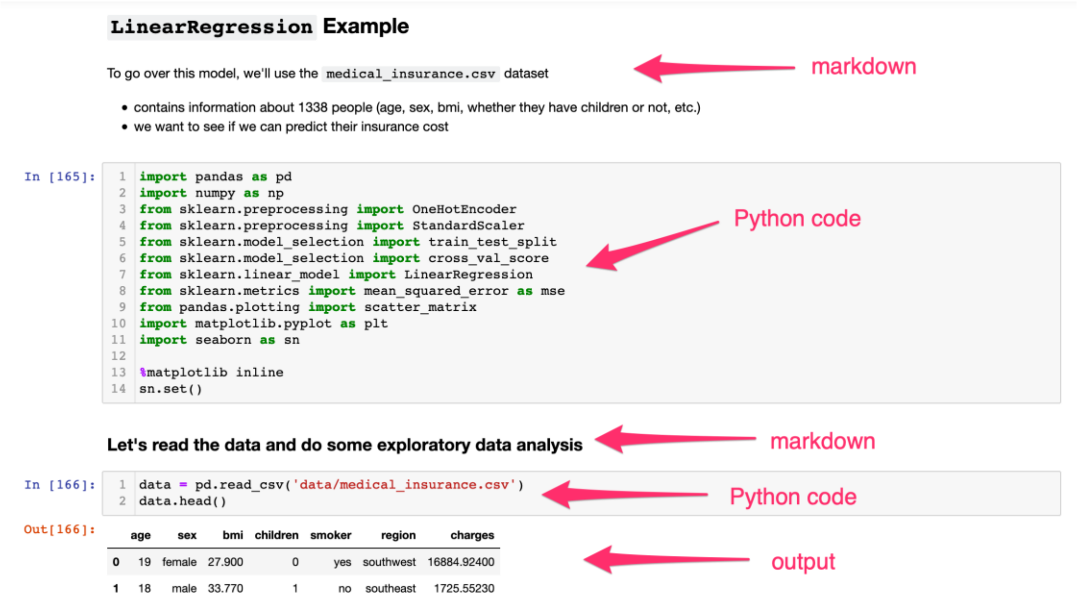
- **The Ecosystem**: Moving from standalone Python scripts to integrated environments like **Jupyter Lab**, **VS Code Notebooks**, and **Google Colab**.
- **Why it matters**: These tools allow for "Literate Programming"—mixing live code, equations, and rich text in one place.
- **Cloud vs. Local**: Using Colab for free GPU/TPU access versus VS Code for heavy local development.

THE VERSION CONTROL CRISIS (NBDIME)

- **The Problem**: Notebooks are stored as complex JSON files. Standard Git "diffs" are unreadable, showing metadata changes instead of code changes.
- **The Solution**: **nbdime** (Notebook Diffing and Merging).
- **Capability**: It allows you to see exactly which code cell or image changed without the "JSON noise".

BRIDGING THE "TECHNICAL GAP" WITH MARKDOWN

- **Stakeholder Communication**: Non-technical managers don't want to read code; they want to read insights.
- **Markdown's Role**: Using headers, bold text, and LaTeX to turn a technical notebook into a professional business report.
- **Executive Summaries**: Every professional notebook should start with a Markdown cell explaining the "Why" and the "Result".



KERNEL MANAGEMENT & REPRODUCIBILITY

- **The "Kernel" Concept**: The engine that executes your code.
- **Version Control**: Understanding how to switch between Python 3.8 and Python 3.11 inside the same interface to avoid library conflicts.
- **Environment Isolation**: Ensuring that your "Agricultural" project doesn't break your "Healthcare" project by using separate kernels.